

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_684a0618-ccb\agent-a63bed7936001a4c5.jsonl`  
- SHA-256: `8275c4d9e2d36b079fa8b1c03c674d48676bd73dbc4ab8ee96b594fa975184ec`  
- Source modified: `2026-06-19T19:18:30+00:00`  
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- project: `wf_684a0618-ccb`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T19:17:29.043Z)

THE IDEA TO ASSESS: a desktop app OR web UI where the user points KhelSutra at an ONLINE storage location they already own/pay for ? a Google Cloud Storage bucket, Google Drive, OneDrive/Microsoft 365, Dropbox, S3, etc. KhelSutra then WATCHES that folder, pulls/streams new recordings as they arrive, and AUTO-prepares highlights (same processing as the local/desktop path). "Bring your own storage + autosync": footage lands in their cloud ? reels appear automatically, zero manual upload.

You are assessing how this idea fits ONE area of the KhelSutra (badminton-highlight-indexer) docs. Read: docs/STORAGE_SHARING_MODEL.md and docs/GOLDEN_DATA_SHARING.md. GOLDEN_DATA_SHARING already uses a OneDrive shared folder that auto-syncs both ways (GPU box ? CPU sessions) ? surface that as an existing precedent for cloud-folder autosync. What does the storage/sharing model say about where user video lives and read-only access?

Be evidence-based and honest ? quote real anchors, don't invent. If the doc does NOT anticipate the idea, say so plainly and note what it WOULD take. Return the structured object.

2. assistant (2026-06-19T19:17:31.598Z)

I'll read both docs to assess how the BYO-storage/autosync idea fits.

3. user (2026-06-19T19:17:32.612Z)

```
1      # Storage-priced sharing ? mental model (DRAFT v2 for co-design)  
2  
3      **Status:** DRAFT ? owner + Claude co-design session pending; nothing here is ratified.  
4      **Date:** 2026-06-12 (v2 adds: model-tier axis, social layer, on-device end-state, abuse  
5      economics ? per owner's second brief) . **Source:** owner briefs (voice note `Umiya  
Velociti.m4a`  
6      pending transcription) + UI_PROPOSAL . HOSTING_PLAN . PMF research . NEXT_STEPS quality  
table.  
7      ? If ratified, this **supersedes the monetization framing in COMMERCIALIZATION.md**  
(away from  
8      per-video/subscription, toward storage-anchored tiers that bundle model quality) ?  
record as an ADR.  
9  
10     ## 1. The one-line model (v2)  
11  
12     Don't sell videos or subscriptions ? **sell shelf space, and let model quality ride the  
shelf**:  
13     free = **2 GB + the owned model**; paid = **more shelf + the best model**; the visible  
meter is  
14     always storage. The social layer makes the shelf worth filling; the owned model makes  
the free  
15     tier cost ~nothing; the end-state moves compute onto the customer's own hardware
```

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entirely.
16
17     ## 1a. The three coupled axes (one user-visible meter)
18
19     | Axis | Free | Paid | End-state (serious users) |
20     |---|---|---|---|
21     | **Storage** (the meter) | 2 GB library | 10 GB+ / pooled | Mostly local disk; cloud =
shareable library only |
22     | **Model** (rides the tier) | Owned model only ? **never Gemini** | Best path (two-
stage refine / frontier) | Best owned model, on-device |
23     | **Compute** (the cost) | Our GPU, capped runs | Our GPU, priority | **Their GPU**
(USB-drive workflow) |
24
25     ## 2. The load-bearing decision: two tiers of object
26
27     | Tier | What | Size reality | Stored? | Quota? |
28     |---|---|---|---|---|
29     | **Source** (raw match upload) | Processing INPUT | 1?4 GB each | Temporarily: kept
through processing + correction window, then **auto-deleted** (e.g. 7 days or on "corrections
confirmed") | Not by default; optional "keep my raw match" toggle = counts |
30     | **Library** (compiled reels, kept rally clips, thumbnails) | The PRODUCT | reel ?
50?80 MB (3?5 min 720p) | Durable | **Yes ? this is what 2 GB means** |
31
32     Why this split: 2 GB cannot hold even one raw match, but holds **~25?40 reels** ? the
quota is
33     only meaningful against outputs. It also kills the "free CDN" abuse vector for a
storage-priced
34     product: **the library only ever contains pipeline outputs** ? there is no path to park
arbitrary
35     files in it. (Sources additionally pass the existing relevance validator.)
36
37     So: **Do we store the uploaded video?** ? **Yes, transiently ? it's a processing input
with a
38     retention deadline, not a possession.** The reel is the possession.
39
40     ## 3. Eviction policies = a pluggable registry (the repo's own pattern)
41
42     `EvictionPolicy` ABC + registry, exactly like `segmenter_registry`:
43     `fifo` (oldest first) . `largest_first` . `smallest_first` . `least_visited` (view-
counter) .
44     `hard_stop` (no deletion ? block new saves until user frees space or upgrades).
45
46     - **Proposed default: `hard_stop`** ? never silently delete someone's match memories;
FIFO etc.
47     are opt-in conveniences. (Co-design question #2.)
48     - **Grace flow:** over quota ? banner + email with the exact list the chosen policy
would evict ?
49     N-day grace (propose 7) ? evict, logged in an `eviction_log`. Offer download-before-
evict.
50     - `least_visited` needs a decision: owner views, public views, or both?
51
52     ## 4. CUJ after upload (end-to-end)
53
54     ```
55     record ? upload source ? async job (PR-1) ? review/correct (corpus engine, PR-4)
56         ? compile reel(s) ? reel lands in LIBRARY (quota meter ticks)
57         ? share: profile page / unlisted link / private
58         ? source hits retention deadline ? auto-delete
59         ? library overflow ? eviction policy (with grace)
60     ```
61
62     A Gmail-style **quota meter** is permanent UI. Profile = `khelsutra.guru@handle`: reel
gallery,
63     view counts (feeds `least_visited`), sport tags. Per-reel visibility: public / unlisted
/ private.

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64
65     ## 5. Upload path: "500 MB cap" vs "point at a bucket"
66
67     - **Alpha (tunnel ? local box):** keep the PR-2 chunked endpoint. A 500 MB cap ? 10?15
min of
68         1080p ? fine for pre-trimmed games, **kills full matches (1?4 GB)**. Recommend keeping
the
69         4 GB chunked cap and rate-limiting per user instead; tomorrow's CUJ run will tell us
real file
70         sizes. (Co-design question #3.)
71     - **Beta (hosted):** the bucket IS the upload path ? browser ? object storage
**directly** via
72         signed resumable URLs (PLATFORM P2 `StorageBackend`); the server never proxies bytes;
"max
73         filesize" becomes policy enforced at URL-issue + post-upload validation, not a proxy
limit.
74
75     ## 6. The real bill is egress, not storage (sharing = bandwidth)
76
77     Storing a free user's 2 GB costs **< ?5/month anywhere**. Serving is the risk: on GCS,
egress is
78     ~$0.12/GB ? one mildly viral reel (80 MB × 1,000 views) ? **$10 of bandwidth**; on
Cloudflare R2,
79     egress is **$0**. ? Proposed hybrid: **sources ? GCS** (credits, lifecycle auto-delete
does the
80     retention for free) . **library/serving ? R2** (zero egress; virality becomes free
marketing
81     instead of a bill). Storage-anchored pricing only has margins on zero-egress serving.
82
83     ## 7. Pricing sketch (INR, storage-anchored ? numbers are placeholders)
84
85     Free **2 GB** ? paid tiers, e.g. +10 GB / +50 GB monthly or yearly. Anchors: Google One
is
86     ?130/mo for 100 GB (we are not selling bytes ? we sell *organized, shareable match
memory* ?
87     priced above raw storage but inside the PMF envelope: B2C $60?180/yr). Academy (B2B
overlay):
88     **pooled storage** across students + coach-managed profiles. Processing cost note:
Gemini path is
89     ~$0.05/run ? decide whether the free tier includes unlimited processing or N
matches/month
90     (quota is storage-anchored, but compute isn't free ? soft cap recommended).
91
92     ## 7a. The model-tier axis ? why "free never touches Gemini" is load-bearing
93
94     1. **Free-tier economics:** Gemini ? $0.05/run forever; the owned model is ~$0 marginal
on owned
95     GPU. A free tier on the owned model is *sustainable at scale*; one on Gemini bleeds.
96     2. **The corpus flywheel becomes the business model:** free users correct rallies (PR-4)
?
97     golden corpus grows ? owned model climbs the quality table (heuristic 0.48 ? Gen-0
0.74 ?
98     target ? wrapper's 0.66 on multi-court) ? free tier literally improves every quarter
?
99     more users ? more corrections. The owned-model roadmap
(OWNED_MODEL_IMPLEMENTATION_PLAN) IS
100     the free-tier product engine.
101     3. **Independence & privacy:** free tier immune to provider 503 waves, price hikes, and
terms
102     drift; paid tier uses Gemini only on its paid/private tier per existing guardrails.
103     4. **Honest quality ladder (from NEXT_STEPS):** free = owned (today Gen-0-grade,
improving);
104     mid = two-stage WASB+refine (0.83 clean); top = best available path (wrapper 0.93
clean).
105     Paid stays ahead until the owned model overtakes ? at which point margins improve

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again.
106
107     ## 7b. Tier sketch (names placeholder ? co-design)
108
109     | | **Rally** (free) | **Smash** (paid) | **Court** (academy/pro) |
110     |---|---|---|---|
111     | Library | 2 GB | 10 GB+ | Pooled across students |
112     | Model | Owned only | Best path + priority queue | Best owned, **on-device** |
113     | Processing | N matches/mo cap | Higher cap | Unlimited (their hardware) |
114     | Social | Full | Full | + coach overlay, athlete profiles |
115
116     ## 7c. Social layer ? "Facebook of (match) videos," scoped honestly
117
118     The shelf is worth paying for only if filling it is social. **Interactions are free;
storage is
119     the meter** ? never paywall the social loop itself.
120     - **v1 social (already in the alpha/beta plan):** profile `@handle`, reel gallery, view
counts,
121     share links. **v2 social (post-PMF, big build):** follows, reactions (badminton-
native: ?
122     "clean smash"), comments, tag-a-player (consent-gated), academy groups, "rally of the
week."
123     - Network effect is local-first: the same court's regulars find each other ? academy =
built-in
124     graph seed.
125     - ? Scope discipline: feed/moderation/notifications is a product in itself; it lands
AFTER the
126     alpha proves retention, not alongside it.
127
128     ## 7d. On-device end-state ? the USB-drive vision
129
130     Serious users plug a drive of match videos into **their** Windows/GPU machine; the app
processes
131     locally (this repo already *is* that pipeline) and syncs only reels + metadata to the
cloud
132     library/profile. Cloud cost for the heaviest users ? storage + social only.
133     - **The licensing guardrail becomes the product moat:** Gemini can never ship on-device;
the
134     owned model (Qwen3-VL base, Apache-2.0 / MIT-only deps) CAN. Commercial-clean
licensing was
135     the prerequisite all along.
136     - Staging: academy pilot boxes first (serious, GPU-equipped, exactly the owner's own
setup) ?
137     consumer Windows app later (needs the WSL-free Linux-native/DirectML detector path ?
the
138     detector ABC seam exists for this).
139     - Maps to PLATFORM P3 `local`/`byo_worker` compute backends ? productized.
140
141     ## 7e. Abuse economics (why "serious pay, abusers can't extract value")
142
143     Each axis is self-defending: **storage** ? library accepts only pipeline outputs (no
arbitrary
144     files), sources auto-expire; **compute** ? free tier = owned model (cheap) + N runs/mo
cap +
145     relevance validator rejects non-sport uploads; **bandwidth** ? R2 zero-egress serving +
signed,
146     revocable links (and the egress allowance baked into paid storage pricing per $6:
price/GB =
147     storage + ?·expected-views·reel-size·egress-rate + margin; on R2 the ?-term ? 0 and
becomes
148     margin); **social** ? report/takedown + rate limits. There is no tier where an abuser
extracts
149     more value than they pay for, and no free capability that costs us marginal money at
scale.
150

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151     ## 8. Guardrails (carry-overs, non-negotiable)
152
153     Consent-at-capture for other players (Phase-2 recording rules) . **minors never on
public
154     profiles** and never in the training pool . report/takedown path on public reels . share
links
155     signed + revocable . training-data opt-in stays separate from storage/sharing consent.
156
157     ## 9. Schema deltas (when we build ? not before ratification)
158
159     `users(quota_bytes, eviction_policy)` . `library_objects(id, owner, video_id,
kind?{reel,clip,thumb},
160     bytes, created_at, last_visited_at, visit_count, visibility, storage_ref)` .
`eviction_log` .
161     source uploads gain `retention_deadline` . `storage_ref` follows P2's `StorageRef`
(backend id + key).
162
163     ## 10. Co-design agenda (the decisions, in order)
164
165     **Object model & policies**
166     1. Ratify the **two-tier model** (sources transient / library durable)?
167     2. **Default eviction:** `hard_stop` (proposed) vs `fifo`? Grace days = 7?
168     3. Quota counts reels only, or also opt-in kept sources? (proposed: both, sources opt-
in)
169     4. `least_visited`: whose visits count?
170
171     **Tiers & pricing**
172     5. Ratify **"free never touches Gemini"** (owned-model-only free tier)?
173     6. Free tier processing cap: N matches/month = ?
174     7. Tier names/sizes (Rally 2 GB / Smash 10 GB / Court pooled ? placeholders) + INR price
points.
175     8. Egress-baked pricing accepted ($7e formula), with R2-for-library hybrid despite GCP
credits?
176
177     **Upload & serving**
178     9. **Alpha upload cap:** 500 MB vs keep 4 GB chunked? (use the first real CUJ file
sizes)
179
180     **Social & end-state**
181     10. v1 social scope = profiles + share links + view counts ONLY (feed/comments post-
PMF)?
182     11. Minors on public profiles: propose **hard exclusion** in v1 (stricter than the
training-pool
183     rule ? public sharing is a different risk class). Ratify?
184     12. Profile namespace: `@handles` on khelsutra.guru?
185     13. On-device milestone placement: academy pilot boxes first, consumer app after the
WSL-free
186     detector path?
187
188     **Paper trail**
189     14. ADR to supersede COMMERCIALIZATION.md's subscription framing once 1?13 settle.
190

```

4. user (2026-06-19T19:17:32.705Z)

```

1     # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3     **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4     **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts
(trajjectory CSVs, `_golden_features.json`, manifests, baseline snapshots) so the **GPU box**
and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6     ## Why this exists
7     The rally-quality measurement is split across two machines by capability:
8     - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the

```

****videos****, the ****trajectory CSVs****, the ****`.rallies.csv` golden labels****, and a ****GPU/WSL**** stack ? only the GPU box can run it.

9 - ****`backend/eval/fusion\_compare.py`** / ****`backend/eval/ablation.py`**** (the LOVO ****litmus****: ?F1, bootstrap CI, paired sign test) are ****pure-CPU re-scoring**** that read only the small ****`\*\_golden\_features.json`** ? any session (incl. a CPU dev container) can run them.**

10

11 Today those artifacts sit in gitignored ****`output/`** (per-machine) and scattered local folders (****`...\Annotation Setup\Collect\Trajectories\`**, ****`...\Golden Labelled\`**). Nothing syncs them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap with OneDrive.

12

13 **## Canonical shared folder**

14 ****`%USERPROFILE%\OneDrive\rally-annotator\golden-data\`** ? ****`C:\Users\avidu\OneDrive\rally-annotator\golden-data\`**

15

16 OneDrive is present + signed-in on both the GPU box and the dev machine under the same account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an overwrite):

17

18 **```**

19 golden-data/

20 README.md

21 features/<corpus-tag>/ ****`\*\_golden\_features.json`** (small ? the key shared artifact)

22 trajectories/<corpus-tag>/ ****`\*\_predict.csv`** / ****`\*\_wasb.csv`** (medium ? reusable across runs)

23 manifests/ ****`\*\_lovo\_manifest.json`**

24 baselines/ leaderboard / regression snapshots (or keep tracked in eval_baselines/)

25 **```**

26

27 ****`<corpus-tag>`** = e.g. ****`2026-06-15-n6`** (date + corpus size). ****Do NOT**** put raw videos here (tens of GB) ? keep those local to the GPU box.

28

29 **## The GPU ? CPU workflow**

30 1. ****GPU box**** (videos + CSVs + labels present):

31 **```**

32 python -m backend.eval.fusion_golden --manifest <manifest> \

33 --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"

34 **```**

35 (or extract to local ****`output/`** then copy the ****`\*\_golden\_features.json`** into the shared folder).

36 2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).

37 3. ****CPU dev/agent session**** points the analysis at the shared folder:

38 **```**

39 python -m backend.eval.fusion_compare --features-dir "...golden-data\features\2026-06-15-n6" --record

40 python -m backend.eval.ablation --features-dir "...golden-data\features\2026-06-15-n6" --granularity group

41 **```**

42 ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's ****`output/`** to also light up the ****`skipif`** golden-litmus tests (****`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`**, ****`tests/test\_fusion\_compare.py`**).

43

44 **## Optional code ergonomics (in #175, not required)**

45 - A ****`RALLY\_GOLDEN\_DIR`** env var (default ****`output/`**) honored by ****`fusion\_golden`** / ****`fusion\_compare`** / ****`ablation`** so neither machine hardcodes the path. (The CLI ****`--out-dir`** / ****`--features-dir`** flags already make this a convention; the env var is just ergonomics.)

46 - A tiny ****`scripts/sync\_golden.py`** to copy ****`output/\*\_golden\_features.json`** ? the shared folder.

47

48 **## Guardrails**

49 - The shared folder lives ****outside the repo**** ? never committed. Keep ****`output/`** gitignored and do ****not**** symlink the shared folder into a tracked path.

50 - The features derive from the **Proprietary** corpus ? keep the OneDrive folder
51 **private** (no public share links).
52 See also: `docs/NEXT\_STEPS.md` (the "Next-session pickup" anchor),
53 `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (P3), `docs/TRACKNET\_WSL\_SETUP.md` (the GPU/WSL detector
setup), and **docs/GOLDEN_REGRESSION_FIXTURES.md** ? the COMMITTED, de-attributed regression
subset (synthetic + owner-cleared) that runs in-repo *without* the videos. This OneDrive
folder stays the **full private corpus**; the committed subset is the minimized/anonymized,
shuttle-only public tier (no person/audio, opaque surrogate ids).

5. assistant (2026-06-19T19:17:35.599Z)

I have both docs read in full. This is a clean, evidence-based assessment ? both files are short enough to quote precisely. Let me compose the structured answer.

6. user (2026-06-19T19:18:17.572Z)

Structured output provided successfully

7. assistant (2026-06-19T19:18:30.869Z)

I've completed the assessment. Summary of what I found across the two docs:

Existing autosync precedent (confirmed): `docs/GOLDEN\_DATA\_SHARING.md` already runs a real cloud-folder autosync ? `%USERPROFILE%\OneDrive\rally-annotator\golden-data\` is "present + signed-in on both the GPU box and the dev machine under the same account, so it auto-syncs," moving GPU-extracted artifacts to CPU sessions "automatically (small files, seconds)," with a `scripts/sync\_golden.py` copy helper. So watch-a-cloud-folder is a lived pattern in the repo ? but it's an *internal, same-account, small-JSON* sync, and its guardrail ("keep the OneDrive folder private ? Proprietary corpus, no public share links") is itself a signal that even internal cloud-folder sync is handled as privacy-sensitive.

Where user video lives / read-only access (the core question):
`docs/STORAGE\_SHARING\_MODEL.md` §2 is explicit ? raw match video is a **Source** tier object: a transient processing INPUT (1?4 GB), "kept through processing + correction window, then auto-deleted," "not a possession." The **Library** (reels) is the durable product and the only thing the 2 GB meter counts. Critically, the doc assumes sources live in **KhelSutra-controlled** storage. The only "point at a bucket" language (§5 Beta) is *our* object storage as the upload TARGET via signed resumable URLs ? a browser-push path, **not** a pull/watch of a folder the customer already owns. There is no read-only "connected external source" concept anywhere.

Verdict: partial anticipation. The idea is the natural autosync front-door to the §7d "BYO storage / their GPU" end-state and slots in as a new **read-only StorageBackend** (PLATFORM P2) + watcher mode feeding the existing §2 source tier and §4 CUJ. What's missing: the watcher/change-feed itself, an OAuth/least-privilege scope model, a "read-only external source" object in the §2/§9 schema, and ? most load-bearing ? an egress/compute-placement decision (pulling 1?4 GB out of the user's cloud into our GPU is the inverse of the §6 serving-egress concern the whole pricing model is built around).

Full evidence, anchors, gaps, and risks are in the structured object.